

Last two classes: NFAs.

Key Mindset: Does this extra feature meaningfully strengthen my computational model?

i.e. Does this feature capture something important?

Another mindset, though: this makes life easier.

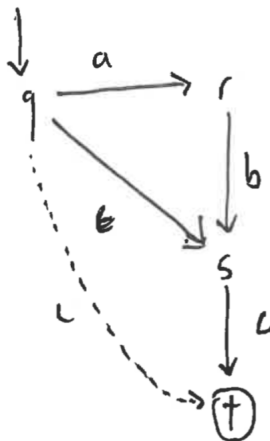
Example: Give DFAs for $A = \{a^k \mid k \equiv 1 \pmod 3\}$. $B = \{b^k \mid k \equiv 2 \pmod 4\}$

To get NFA for $A \cup B$: just use multiple start states!

Epsilon Transitions

Idea: State changes without reading characters.

Example:



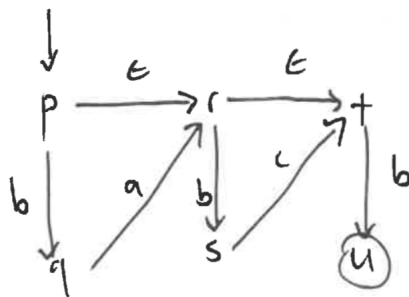
abc is accepted because of non-epsilon path.

But at the start, we could take the epsilon transition.

$\Rightarrow c$ is also accepted.

Language: $\{c, abc\}$.

Example:



What are the accepting paths for each of the strings in the language?

Theorem: Any language that has an ϵ -NFA has an NFA.

(convenient, and insightful! wow.)

Concatenation and Asterate

Theorem: The set of regular languages is closed under concatenation and asterate.

Recall: Given two languages A and B , $AB = \{xy \mid x \in A, y \in B\}$

Example: $A = \{a^k \mid k \equiv 0 \pmod{3}\}$. $B = \{b^k \mid k \equiv 0 \pmod{4}\}$

Draw NFAs for:

- AB
- $(\sim A)(\sim B)$

General approach for concatenate. Let M_A and M_B be NFAs for A and B respectively.

- Attach all final states of M_A to all start states of M_B by ϵ -transitions.
- Un-finalize all final states of M_A . Un-start-state-ize all start states of M_B .

Recall: $A^* = A^0 \cup A^1 \cup A^2 \cup A^3 \cup \dots$

Ex: $A^* = \{a, bb\}^* = \{\epsilon, a, bb, aa, abb, bba, bbbb, \dots\}$

What if we take NFA for A and add a loop back from final states?

Missing ϵ ! I.e. we've got $A^1 \cup A^2 \cup A^3 \cup \dots$

Add NFA for $\{\epsilon\}$. Union!

Removing Epsilon Transitions

Idea: Replace ϵ -transitions with real transitions. Then remove ϵ -transitions.

Example: $\{c, abc\}$

Idea: replace $p \rightarrow_\epsilon q \rightarrow_c r$ with $p \rightarrow r$.

Let's try with $\{babcb, bab, bcb, b\}$

$\{babcb, bcb\} \dots$ oops.

Issue: Chains of ε -transitions.

Def: Given a state q , $\text{eps}(q)$ is the set of states reachable from q using only chains of ε transitions.

For all $q, p \in \text{eps}(q)$ and $p \rightarrow_c r$, add $q \rightarrow_c r$.

Now remove epsilon transitions.

Thing to think about: what should the start/final states be now?